CS 340 Project Two – Grazioso Salvare Dashboard

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Project Overview

This project is a full-stack dashboard application developed for Grazioso Salvare, an organization that trains rescue dogs. The purpose of the application is to allow users to interact with and visualize animal shelter data in order to identify dogs suitable for different types of rescue operations.

The system integrates a MongoDB database with a Python-based dashboard built using the Dash framework. The application follows the Model-View-Controller (MVC) design pattern, where MongoDB serves as the model, Dash components serve as the view, and Python callbacks act as the controller.

Project Functionality

The dashboard includes the following features:

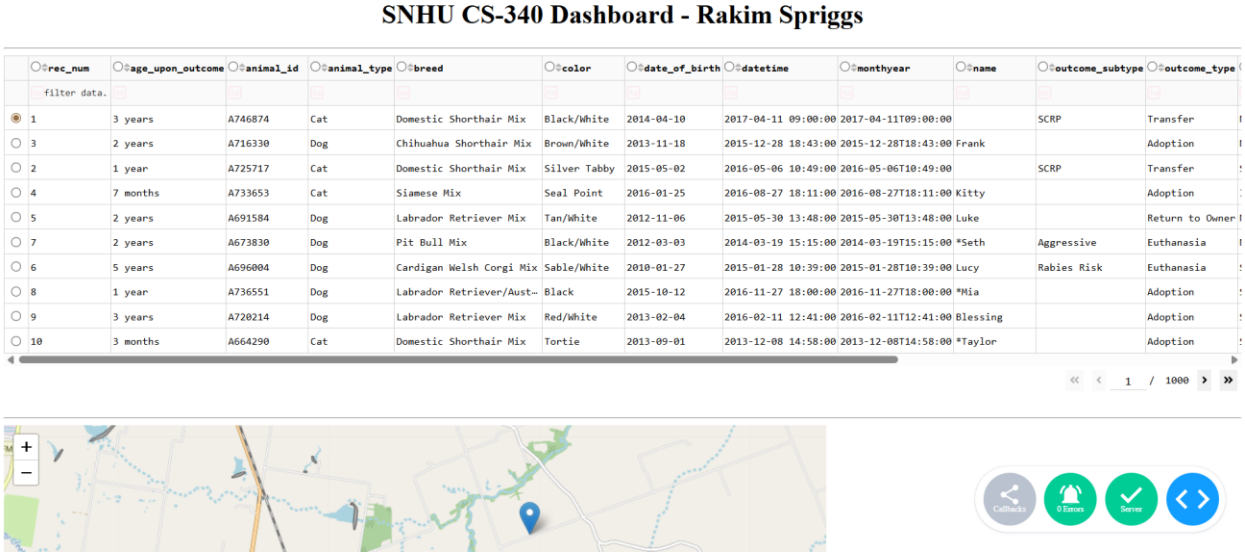
- Interactive data table displaying animal shelter data
- Filtering options for:
    - Water Rescue
    - Wilderness Rescue
    - Disaster Rescue
    - Reset (All data)
- Dynamic updates to:
    - Data table
    - Geolocation map
    - Data visualization chart
- Geolocation map displaying animal locations
- Chart visualizing breed distribution

- Grazioso Salvare logo and user identifier

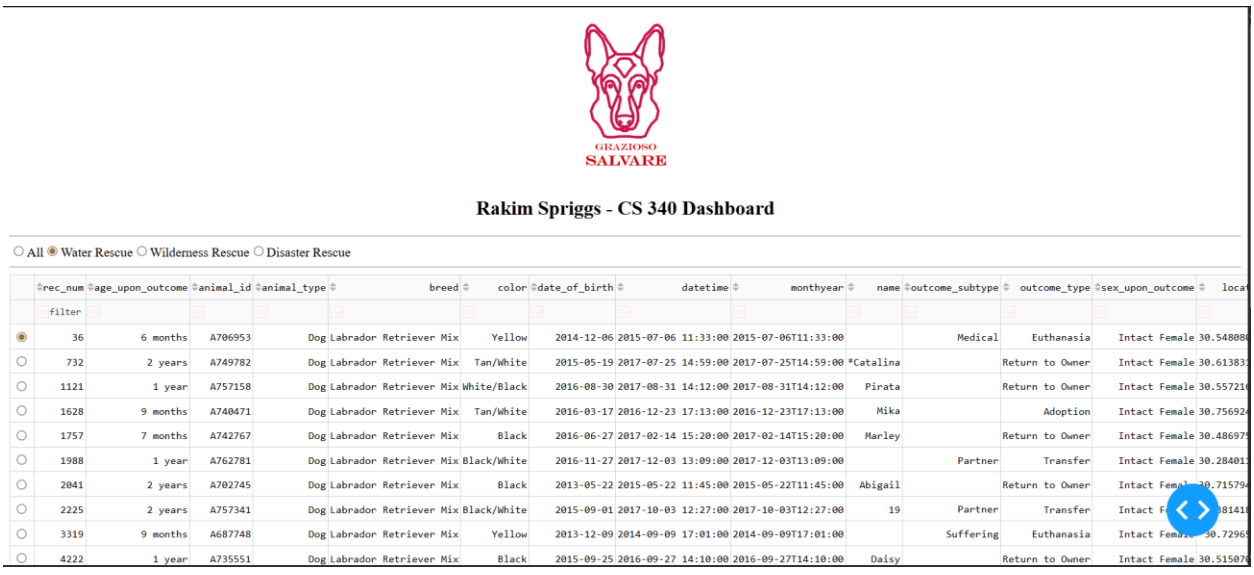
Dashboard Screenshots (Required)

Insert screenshots below:

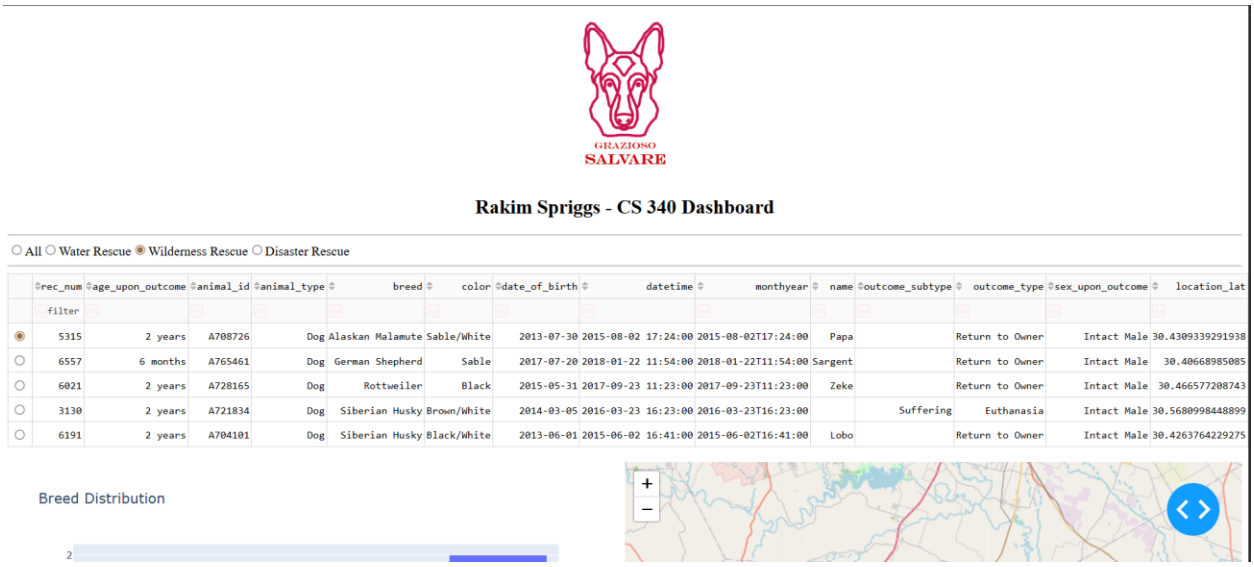
1. Default Dashboard View



2. Water Rescue Filter Applied

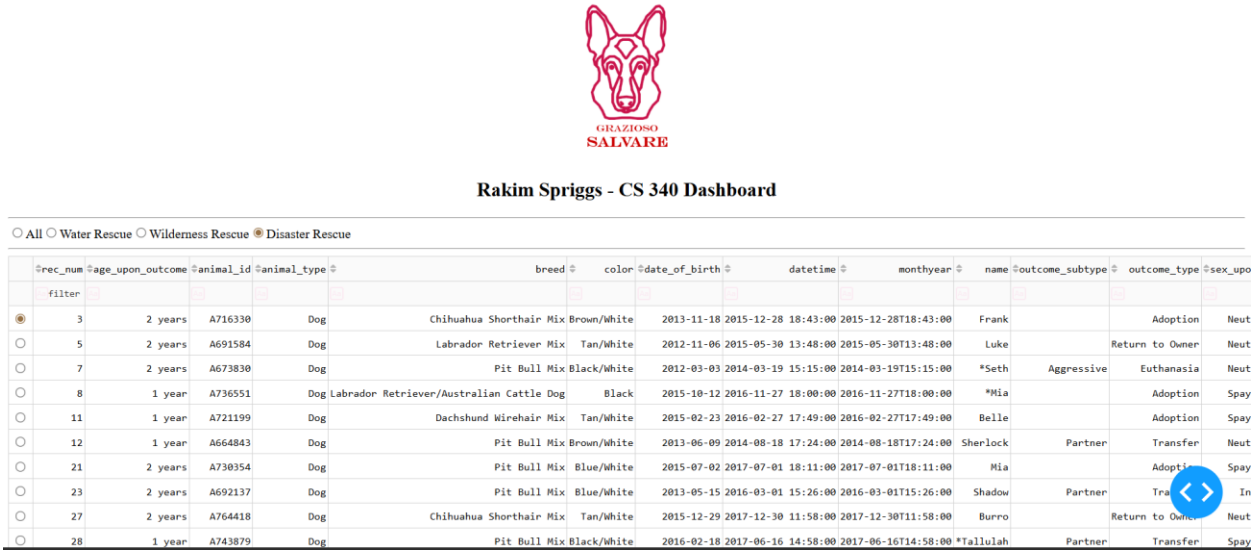


3. Wilderness Rescue Filter Applied



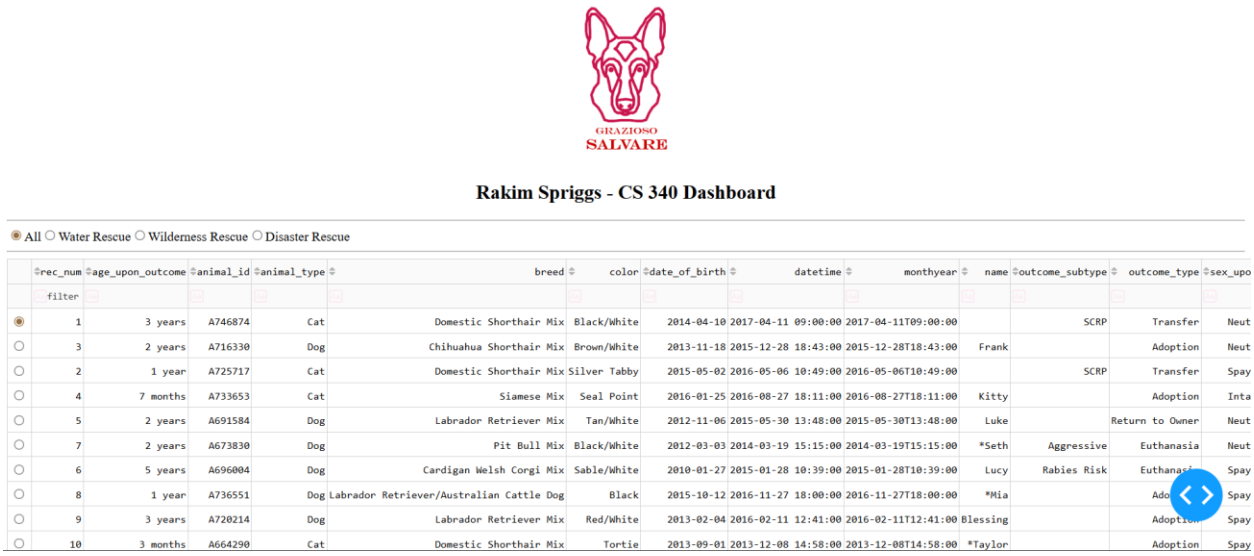
	rec_num	age_upon_outcome	animal_id	animal_type	breed	color	date_of_birth	datetime	monthyear	name	outcome_subtype	outcome_type	sex_upon_outcome	location_lat
	filter													
◉	5315	2 years	A708726	Dog	Alaskan Malamute	Sable/White	2013-07-30	2015-08-02 17:24:00	2015-08-02T17:24:00	Papa		Return to Owner	Intact Male	30.4309339291938
○	6557	6 months	A765461	Dog	German Shepherd	Sable	2017-07-20	2018-01-22 11:54:00	2018-01-22T11:54:00	Sargent		Return to Owner	Intact Male	30.4066898085
○	6021	2 years	A728165	Dog	Rottweiler	Black	2015-05-31	2017-09-23 11:23:00	2017-09-23T11:23:00	Zeke		Return to Owner	Intact Male	30.466577208743
○	3130	2 years	A721834	Dog	Siberian Husky	Brown/White	2014-03-05	2016-03-23 16:23:00	2016-03-23T16:23:00		Suffering	Euthanasia	Intact Male	30.5680998448899
○	6191	2 years	A704101	Dog	Siberian Husky	Black/White	2013-06-01	2015-06-02 16:41:00	2015-06-02T16:41:00	Lobo		Return to Owner	Intact Male	30.4263764229275

4. Disaster Rescue Filter Applied



	rec_num	age_upon_outcome	animal_id	animal_type	breed	color	date_of_birth	datetime	monthyear	name	outcome_subtype	outcome_type	sex_upo
	filter												
◉	3	2 years	A716330	Dog	Chihuahua Shorthair Mix	Brown/White	2013-11-18	2015-12-28 18:43:00	2015-12-28T18:43:00	Frank		Adoption	Neut
○	5	2 years	A691584	Dog	Labrador Retriever Mix	Tan/White	2012-11-06	2015-05-30 13:48:00	2015-05-30T13:48:00	Luke		Return to Owner	Neut
○	7	2 years	A673830	Dog	Pit Bull Mix	Black/White	2012-03-03	2014-03-19 15:15:00	2014-03-19T15:15:00	*Seth	Aggressive	Euthanasia	Neut
○	8	1 year	A736551	Dog	Labrador Retriever/Australian Cattle Dog	Black	2015-10-12	2016-11-27 18:00:00	2016-11-27T18:00:00	*Mia		Adoption	Spay
○	11	1 year	A721199	Dog	Dachshund Wirehair Mix	Tan/White	2015-02-23	2016-02-27 17:49:00	2016-02-27T17:49:00	Belle		Adoption	Spay
○	12	1 year	A664843	Dog	Pit Bull Mix	Brown/White	2013-06-09	2014-08-18 17:24:00	2014-08-18T17:24:00	Sherlock	Partner	Transfer	Neut
○	21	2 years	A730354	Dog	Pit Bull Mix	Blue/White	2015-07-02	2017-07-01 18:11:00	2017-07-01T18:11:00	Mia		Adopti	Spay
○	23	2 years	A692137	Dog	Pit Bull Mix	Blue/White	2013-05-15	2016-03-01 15:26:00	2016-03-01T15:26:00	Shadow	Partner	Tra	In
○	27	2 years	A764418	Dog	Chihuahua Shorthair Mix	Tan/White	2015-12-29	2017-12-30 11:58:00	2017-12-30T11:58:00	Burro		Return to Owne	Neut
○	28	1 year	A743879	Dog	Pit Bull Mix	Black/White	2016-02-18	2017-06-16 14:58:00	2017-06-16T14:58:00	*Tallulah	Partner	Transfer	Spay

5. Reset (All Data)



Tools and Technologies Used

- **MongoDB**
  - NoSQL database used to store animal shelter data
  - Flexible document-based structure
  - Efficient querying for large datasets
- **Python**
  - Used to build application logic and database interaction
  - Integrates with MongoDB via PyMongo
- **Dash Framework**
  - Used to create the web-based dashboard
  - Provides interactive components and callback system
- **Pandas**
  - Used for data manipulation and transformation
- **Plotly Express**
  - Used to create dynamic charts and visualizations
- **Dash Leaflet**
  - Used to generate geolocation maps

Why MongoDB Was Used

MongoDB was chosen because it:

- Stores data in JSON-like documents, making it flexible and easy to work with
- Integrates seamlessly with Python
- Supports efficient querying and filtering
- Handles large datasets effectively
- Allows dynamic schema changes

How the Project Was Completed

1. Imported the Austin Animal Center dataset into MongoDB
2. Created a CRUD Python module to interact with the database
3. Developed a dashboard using Dash
4. Connected the dashboard to MongoDB using the CRUD module
5. Implemented filtering logic based on rescue categories
6. Created interactive components (table, chart, and map)
7. Tested functionality and captured screenshots

Challenges and Solutions

Challenge: MongoDB authentication issues
Solution: Identified that authentication was disabled and adjusted connection to use localhost without credentials

Challenge: Module import errors
Solution: Created and properly configured the CRUD Python module

Challenge: Dash layout syntax errors
Solution: Fixed bracket mismatches and ensured proper layout structure

Challenge: Logo not displaying
Solution: Used base64 encoding to correctly load the image in Dash

How to Run the Project

1. Ensure MongoDB is running
2. Import the dataset into the `aac` database
3. Place the logo file in the correct directory (e.g., `/assets`)
4. Open the Jupyter Notebook file
5. Run all cells
6. Launch the dashboard using:

```
app.run_server(port=8051)
```

Conclusion

This project demonstrates the successful integration of database systems with a client-facing dashboard application. By using MongoDB and Dash, the application allows users to efficiently filter, visualize, and analyze rescue animal data. The final product is a fully functional, interactive dashboard that meets the requirements of Grazioso Salvare.

Resources

- MongoDB Documentation: https://www.mongodb.com/docs/
- Dash Documentation: https://dash.plotly.com/
- Plotly Express: https://plotly.com/python/plotly-express/
- Pandas Documentation: https://pandas.pydata.org/